

ABID UR REHMAN

BS Biotechnology | Molecular Diagnostics · Bioinformatics · Nanobiotechnology

Chitral, KPK, Pakistan • +92 344 1434341 • abidurehman41d@gmail.com

[linkedin.com/in/abid-ur-rehman](https://www.linkedin.com/in/abid-ur-rehman)

PROFESSIONAL SUMMARY

Final-year BS Biotechnology student (CGPA 3.93) and published researcher with hands-on expertise spanning the wet lab and the dry lab. Skilled in molecular diagnostics (PCR/RT-PCR, ELISA, serology), bioinformatics (BLAST, Clustal Omega, phylogenetics, resistance-gene mapping), and green nanotechnology. Recipient of a competitive ORIC research grant and co-author of a peer-reviewed publication. Driven to apply biotechnological innovation to real-world problems in health and the environment.

EDUCATION

BS Biotechnology — Capital University of Science & Technology (CUST), Islamabad 2022 – 2026 (Expected)

- CGPA: 3.93/4.00 • 2× Chancellor's Honor Roll • 5× Dean's Honor Roll

HSSC (Pre-Medical) — Aga Khan Higher Secondary School, Chitral 2021

- 88.7%

SSC — Aga Khan Higher Secondary School, Chitral 2019

- 84.5%

RESEARCH & PUBLICATIONS

Co-Author — Peer-Reviewed Publication Planta Animalia, Vol. 5

- "Detection and Confirmation of Candida Species in Stored Apples from Diverse Regions of Punjab." Conducted PCR amplification using ITS1/ITS4 primers and species confirmation.

Final-Year Research Project (ORIC-Funded) 2025 – 2026

Green-Synthesized Silica Nanoparticles from Cuscuta reflexa Extract

- Secured competitive funding through the CUST FYP Funding Program (COHORT-6) — one of only six projects selected by the Office of Research, Innovation & Commercialization (ORIC).
- Green synthesis of silica nanoparticles evaluated for antimicrobial and antilarval activity, plus in-vivo toxicological assessment for biomedical safety and biocompatibility.

In-Silico Genomics Research

- Mapped chromosomal integration and genetic regulatory switches of antibiotic-resistance genes in Acinetobacter baumannii using BPROM and ResFinder.

Book Chapter (In Progress)

- Lead contributor on "Medicinal Plants of African Origin for Nanoparticle Synthesis: Antimicrobial and Antilarval In-Vitro Biological Activities."

EXPERIENCE

Laboratory Intern — District Headquarters Hospital, Chitral 2025

- Performed clinical sample handling, diagnosis, and laboratory quality-control protocols.
- Gained hands-on experience in routine diagnostics (CBC, LFTs, KFTs), serological testing, and molecular detection workflows.

Academic Tutor — Islamabad School & Academy, Sihala 2025 – 2026

- Designed conceptual examinations and MCQ assessments aligned to Federal Board curricula (grades 9–12).

SEO Content Writer & Scientific Blogger — Zee Tech Company 2021 – 2023

- Developed SEO-driven content strategies and authored scientific blog content for general audiences.

Mentor — Empower You Camp, Sharar Organization

2023

- Delivered mentorship and personal-development sessions to youth participants.

Volunteer — Street Children Welfare Foundation, Islamabad

2025 – Present

- Volunteering across Poverty Alleviation Programs, Zakat Distribution, Safe Drinking Water Projects, and Livelihood Projects, handling assigned responsibilities diligently.

Volunteer — Helping Hand for Relief & Development, Chitral

2025

- Contributed to community relief and development initiatives for underserved populations.

TECHNICAL SKILLS

Molecular Techniques: PCR, RT-PCR, gel electrophoresis, DNA/RNA extraction, plasmid isolation, restriction digestion, microbial culturing, spectrophotometry, aseptic technique.

Molecular & Clinical Diagnostics: Viral detection (HBV, HCV, HPV), ELISA & serological testing, blood grouping (ABO, Rh), CBC, LFTs, KFTs.

Bioinformatics (Dry Lab): BLAST, Clustal Omega, sequence alignment, phylogenetic analysis, genome annotation, BPROM, ResFinder; Python for biological data analysis.

Biochemical Assays: Protein estimation (Bradford, Biuret), enzyme activity assays, quantitative biomolecule tests, buffer/pH preparation, titration.

Cell Biology: Mitosis/meiosis observation, karyotyping, microscopy, gene-expression analysis, slide preparation.

Communication: Scientific & academic writing, literature review, curriculum design, digital science storytelling.

HONORS & AWARDS

- ORIC Research Grant Winner — CUST FYP Funding Program, COHORT-6 (1 of 6 selected).
- Teach For Pakistan Fellow — selected for Phase 1 of the competitive national fellowship.
- 2× Chancellor's Honor Roll & 5× Dean's Honor Roll, CUST.
- 2nd Position, Poster Presentation (Marfan Syndrome), World DNA Day, CUST.
- Batch Coordinator — elected representative of Batch 223 for two academic years.
- Departmental Football Captain (Bioscience & Bioinformatics) & District Football Association (Chitral Lower) player.

LEADERSHIP & ACTIVITIES

- Core Member, Liaison Team — Youth Empowerment Club, CUST.
- Co-Founder — BioHackers, a digital platform translating genetic-engineering concepts for public science literacy.
- Teach-a-Child Program — Aga Khan Education Service, Pakistan.

CERTIFICATIONS & RECOGNITION

- Winner, Poster Presentation — GENIX Society, CUST.
- Core Member, Liaison Team — Youth Empowerment Club, CUST.
- Mentor, Empower You Camp — SHARAR Organization.
- Mentorship Session on Freelancing — Youth Empowerment Program, CUST.
- Community Service, Teach-a-Child Program — Aga Khan Development Network (AKDN), Pakistan.
- Inter-University Debate Competition — CALC, CUST.
- Executive Volunteer Symposium — Youth Empowerment Club, CUST.
- Inter-Departmental Football Tournament — CUST (2023–2025).
- Youm-e-Iqbal Bait Baazi (Literary) Competition — Capital Arts & Literary Club, CUST.

LANGUAGES

English (Fluent) • Urdu (Fluent) • Chitrali / Khowar (Fluent) • Pashto (Intermediate)